



Name: _____

Student Number: _____

2025

**YEAR 12 MATHEMATICS EXTENSION 2
TASK 1**

Mathematics

**General
Instructions**

- Reading time – 10 minutes
- Working time – 90 minutes
- Write using black or blue pen
- Diagrams are not to scale
- NESA Approved calculators may be used
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:
55**

Section I – 5 marks (page 1–2)

- Attempt Questions 1–5
- Allow about 10 minutes for this section

Section II – 45 marks (pages 3–7)

- Attempt Questions 6–8
- Allow about 1 hour and 20 minutes for this section

Two-sided A4 handwritten sheet 5 marks

Section I**5 marks****Attempt Questions 1–5****Each question in this section is worth 1 mark****Allow about 10 minutes for this section**

Use the multiple-choice answer sheet for questions 1-5.

Question 1Which of the following is equivalent to $-3 - 3\sqrt{3}i$?

- A. $6e^{-\frac{\pi}{3}i}$
- B. $6e^{-\frac{2\pi}{3}i}$
- C. $6e^{\frac{\pi}{3}i}$
- D. $6e^{\frac{2\pi}{3}i}$

Question 2The sum of the roots of $z^3 - 5z^2 + 11z - 7 = 0$, where $z \in \mathbb{C}$, is

- A. 5
- B. $5i$
- C. $2 + i$
- D. $2 - i$

Question 3Given $z = 1 - i$ which expression is equal to z^5 ?

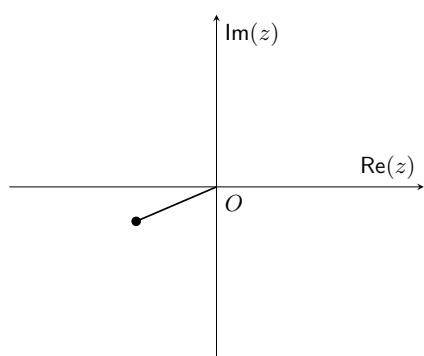
- A. $z^5 = 4\sqrt{2} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$
- B. $z^5 = 4\sqrt{2} \left(\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right) \right)$
- C. $z^5 = 4\sqrt{2} \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$
- D. $z^5 = 4\sqrt{2} \left(\cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right) \right)$

Question 4For what values of n is $(1 + \sqrt{3}i)^n$ purely real?

- A. Multiples of 2
- B. Multiples of 3
- C. Multiples of 4
- D. Multiples of 6

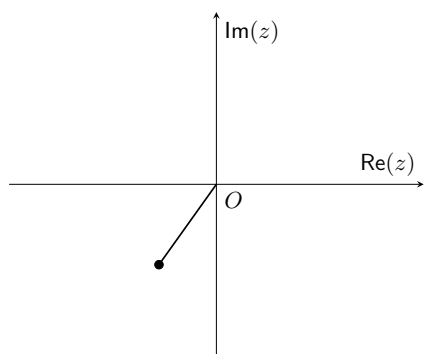
Question 5

The Argand diagram shows the complex number $e^{i\theta}$.

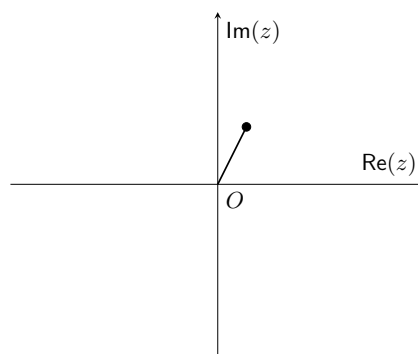


Which of the following could be the complex number $ie^{-i\theta}$?

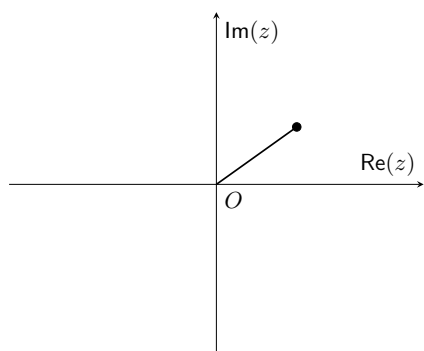
A.



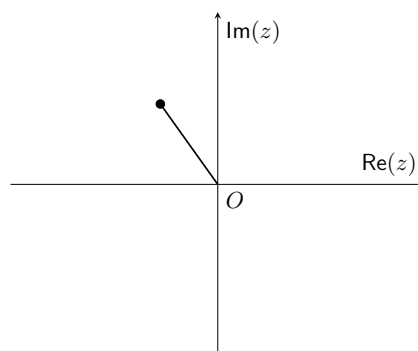
C.



B.



D.



Section II

45 marks

Attempt Questions 6 –8

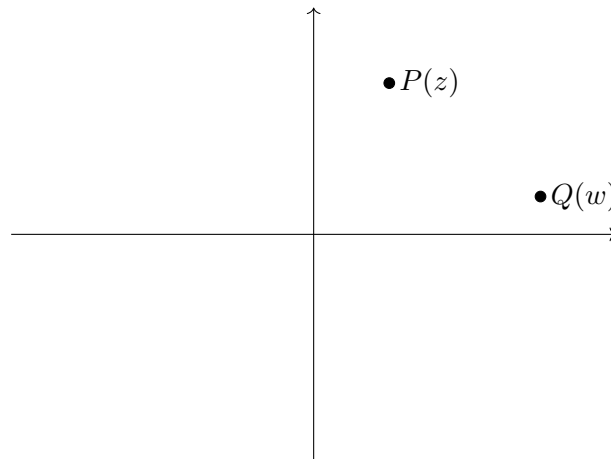
Allow about 1hrs and 20 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 marks)	Marks
(a) Consider the complex numbers $w = 1 + 2i$ and $z = 3 - i$. Find in the form $x + iy$,	
(i) $\overline{w}z$,	2
(ii) $\overline{\left(\frac{10}{z}\right)}$.	2
(b) Factorise $z^4 - z^2 - 20$ over:	
(i) the real field,	1
(ii) the complex field.	1
(c) (i) Find the square roots of $-8 + 6i$ in Cartesian form,	2
(ii) Hence solve $z^2 - (7 - i)z + (14 - 5i) = 0$.	2
(d) Find a quadratic equation in the form $az^2 + bz + c = 0$, where $a, b, c \in \mathbb{R}$ that has the complex root $\sqrt{3} + 5i$.	2

- (e) The points P and Q on the Argand diagram represent the complex numbers z and w respectively.



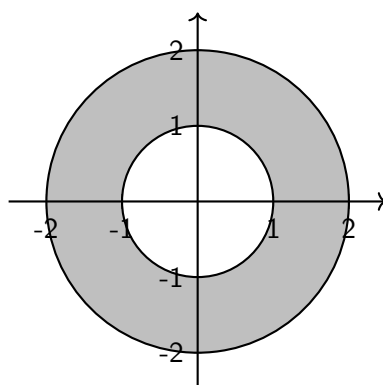
On the **separate answer sheet** provided for Question 6(e) mark on it the following points:

- | | |
|--|----------|
| (i) the point R representing $\bar{z}$, | 1 |
| (ii) the point S representing $z + w$, | 1 |
| (iii) the point T representing $z - w$. | 1 |

End of Question 6

Question 7 (15 marks)**Marks**(a) Let $\omega = -\sqrt{3} + i$.(i) Write ω in modulus-argument form,**2**(ii) Hence find ω^9 in Cartesian form.**1**

(b) The diagram below shows the shaded region in the Argand diagram. Find the inequality which describes the shaded area.

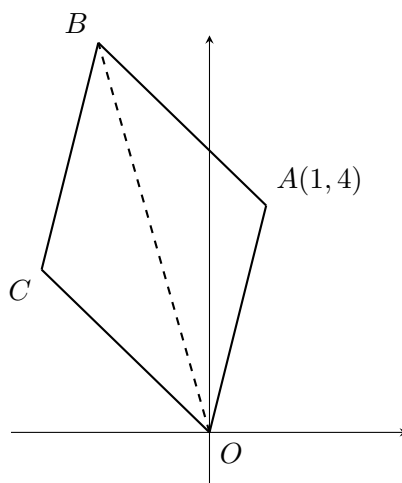
1(c) If p , q and r are the roots of $z^3 + (1 - 5i)z^2 + (-5 + 2i)z + 15 - 3i = 0$, find:(i) $\bar{p} + \bar{q} + \bar{r}$,**2**(ii) $\bar{p} \times \bar{q} \times \bar{r}$.**2**(d) Find the complex roots of $z^6 = -1$ and express them in exponential form.**2**

(e) Sketch the graph of

2

$$\text{Arg} \left[\frac{z - (2 + i)}{z - (1 + 2i)} \right] = \pi$$

- (f) $OABC$ is a rhombus in the Argand diagram, where O is the origin and point B is in the second quadrant. The point A is $(1, 4)$ and the angle AOB is $\frac{\pi}{6}$. **3**



Find the complex numbers represented by the points B and C in Cartesian form.

End of Question 7

Question 8 (15 marks)**Marks**

- (a) The complex number z moves such that $\operatorname{Im}\left[\frac{1}{\bar{z}-i}\right] = 1$. Find the Cartesian equation that describes the locus of z . **3**

- (b) Sketch the intersection of the two regions below: **3**

$$|z + 1 - i| < \sqrt{2}$$

$$-\pi \leq \operatorname{Arg}(z + 1 - i) \leq \frac{3\pi}{4}$$

in the complex number plane.

- (c) Given $|z| < \frac{1}{3}$, show that $|(1+i)z^3 + iz| < \frac{4}{9}$. **3**

- (d) (i) Use the binomial theorem to expand $(\cos(\theta) + i\sin(\theta))^5$, **1**

- (ii) Use de Moivre's theorem and your results from part (i) to prove that **1**

$$\sin(5\theta) = \sin^5(\theta) - 10\cos^2(\theta)\sin^3(\theta) + 5\cos^4(\theta)\sin(\theta),$$

- (iii) Hence or otherwise prove, **2**

$$\tan(5\theta) = \frac{5\tan(\theta) - 10\tan^3(\theta) + \tan^5(\theta)}{1 - 10\tan^2(\theta) + 5\tan^4(\theta)},$$

- (iv) Hence show that $x^4 - 10x^2 + 5 = 0$ has roots $x = \pm \tan \frac{\pi}{5}$ and $x = \pm \tan \frac{2\pi}{5}$. **2**

End of Question 8

End of Task

Name _____

Teacher _____

2025 HSC Mathematics Extension 2

Task 1 Examination

Section I – Multiple Choice

Allow about 10 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you have changed your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word **correct** and drawing an arrow as follows:

A ☐ B ☒ C ☐ D ☐
↑
correct

1. A ☐ B ☒ C ☐ D ☐

2. A ☒ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☒ D ☐

4. A ☐ B ☒ C ☐ D ☐

5. A ☒ B ☐ C ☐ D ☐

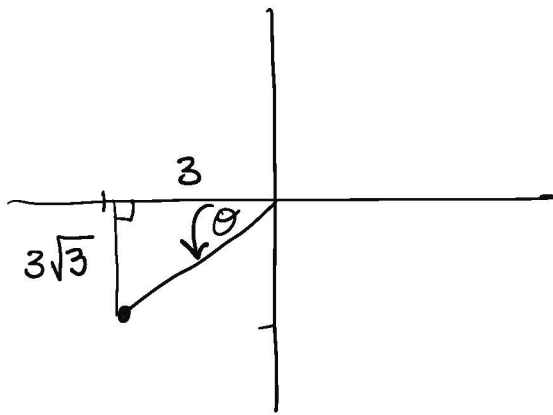
$$1. \quad -3 - 3\sqrt{3}i$$

$$\text{let } z = -3 - 3\sqrt{3}i$$

$$|z| = \sqrt{(-3)^2 + (3\sqrt{3})^2}$$

$$= \sqrt{36}$$

$$= 6$$



$$\arg(z) = -\pi + \tan^{-1}\left(\frac{3\sqrt{3}}{3}\right)$$

$$= -\pi + \frac{\pi}{3}$$

$$= -\frac{2\pi}{3}$$

$$\therefore -3 - 3\sqrt{3}i = 6e^{-\frac{2\pi}{3}i}$$

(B)

2. $z^3 - 5z^2 + 11z - 7 = 0$

let α, β and γ be the roots

let $\alpha = a + bi$

$\beta = a - bi$

$\alpha + \beta + \gamma = \frac{-b}{a}$

$2a + \gamma = +5$

note. 3rd root, as it must be real.

$\therefore$ (A) is the answer.

test $z = 1$

$1 - 5 + 11 - 7 = 0$

$-12 + 12 = 0$

$\therefore \gamma = 1$

$2a + 1 = 5$

$2a = 4$

$a = 2$

$\therefore$ the other roots

$2 + i, 2 - i$

sum of roots.

$1 + 2 + i + 2 - i = 5,$

3. $z = 1 - i$

$|z| = \sqrt{2}$

$\arg(z) = -\frac{\pi}{4}$

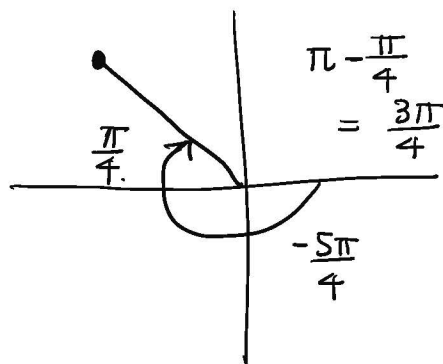
$z = \sqrt{2} \cos -\frac{\pi}{4}$

$z^5 = (\sqrt{2} \cos -\frac{\pi}{4})^5$

$z^5 = (\sqrt{2})^5 \cos -\frac{5\pi}{4}$

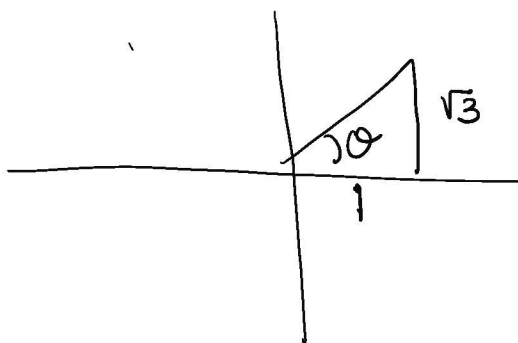
from the diagram

$z^5 = 4\sqrt{2} \cos \frac{3\pi}{4}$



(C)

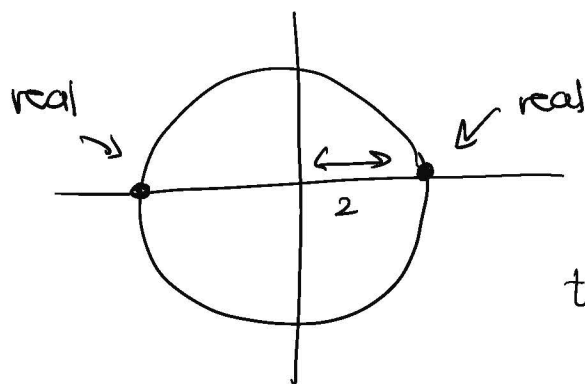
Q4



$$(1 + \sqrt{3}i)^n$$

$$|z| = 2$$

$$\arg(z) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{\pi}{3}$$



$$(1 + \sqrt{3}i)^n = (2 \cos \frac{\pi}{3})^n = 2^n \cos \frac{n\pi}{3}$$

to be real

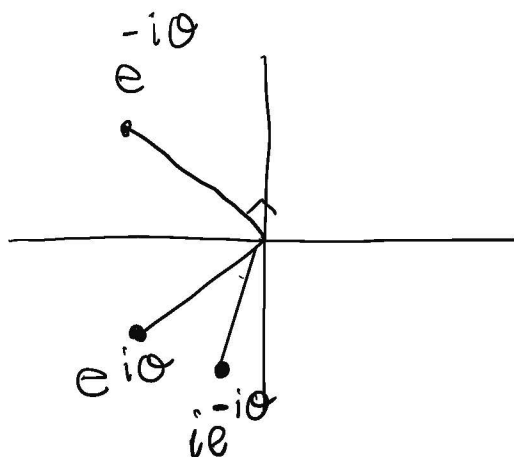
$$\frac{n\pi}{3} = k\pi \text{ where } k \in \mathbb{Z}$$

$$n = 3k$$

i.e. its a multiple of 3.

(B)

Q5



(A)

Question 6

a) $w = 1 + 2i$

(i) $\bar{w}z = (1 - 2i)(3 - i)$ 1 mark correct conjugate

$$= 3 - i - 6i - 2$$

$$= 1 - 7i$$

1 mark for correct solution.

(ii) $\overline{\left(\frac{10}{z}\right)} = ?$

$$\left(\frac{10}{z}\right) = \frac{10}{3-i} \times \frac{3+i}{3+i}$$

1 mark, realising the denominator correctly

$$= \frac{30 + 10i}{9 + 1}$$

$$= \frac{\cancel{10}(3+i)}{\cancel{10}}$$

$$= 3 + i$$

$$\overline{\left(\frac{10}{z}\right)} = 3 - i$$

1 mark correct solution

Question 6.

(b) $z^4 - z^2 - 20 = (z^2 - 5)(z^2 + 4)$ rational field

real field $= (z - \sqrt{5})(z + \sqrt{5})(z^2 + 4)$

complex field $= (z - \sqrt{5})(z + \sqrt{5})(z - 2i)(z + 2i)$

1 mark for each correct solution

(c) (i) $-8 + 6i$

$$\frac{6}{2} = 3 \begin{matrix} 3, 1 \\ 1, 3 \end{matrix}$$

$$1^2 - 3^2 = -8$$

$\therefore$ the square roots are $\pm (1 + 3i)$ 2mark.

(ii) $z^2 - (7-i)z + (14-5i) = 0$

$$\begin{aligned} z &= \frac{7-i \pm \sqrt{(7-i)^2 - 4 \times 1 \times (14-5i)}}{2} \quad -i \times -i \\ &= \frac{7-i \pm \sqrt{49 - 14i - 1 - 56 + 20i}}{2} \\ &= \frac{7-i \pm \sqrt{-8+6i}}{2} \end{aligned}$$

$$\therefore z = \frac{7-i + 1+3i}{2}$$

$$\begin{aligned} &= \frac{8+2i}{2} \\ &= 4+i \quad \checkmark \text{ 1mark} \end{aligned}$$

$$\text{or } \frac{7-i - 1-3i}{2}$$

$$\begin{aligned} &= \frac{6-4i}{2} \quad \swarrow \text{1mark} \\ &= 3-2i \quad \searrow \text{1mark} \end{aligned}$$

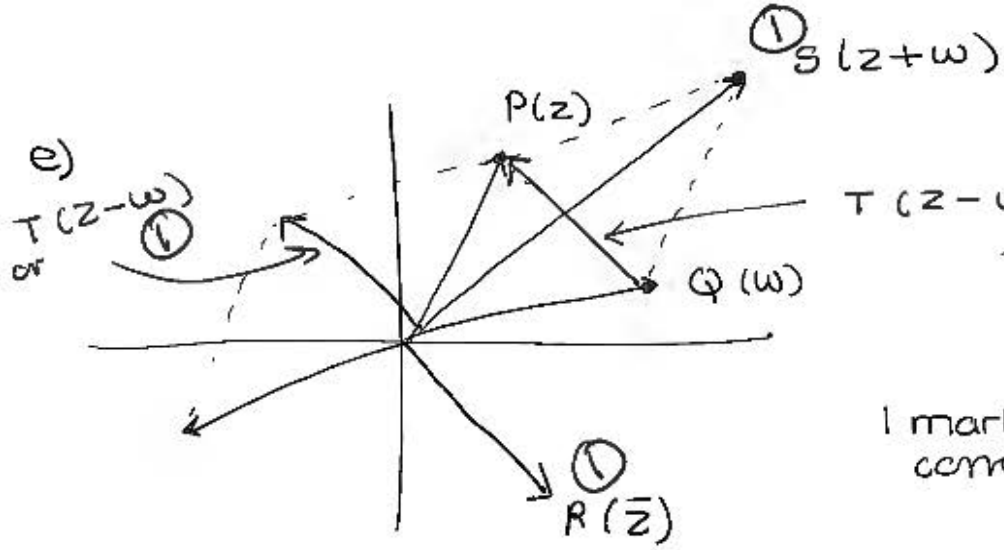
d) $(z - (\sqrt{3} + 5i))(z - (\sqrt{3} - 5i)) = 0$

$$z^2 - 2(\sqrt{3} - 5i)z + (\sqrt{3} + 5i)(\sqrt{3} - 5i) = 0$$

$$z^2 - 2\sqrt{3}z + (3+25) = 0$$

$$z^2 - 2\sqrt{3}z + 28 = 0$$

1mark for correct solution



that's
the vector
the point
is the position
vector

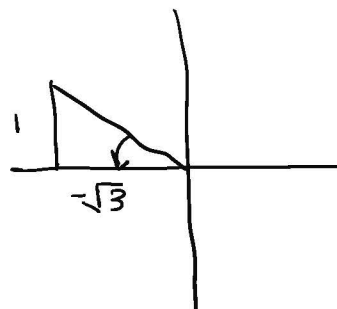
1 mark for each
correct solution

Q7. Simplify

a) (i) $\omega = -\sqrt{3} + i$

$$|\omega| = \sqrt{(-\sqrt{3})^2 + (1)^2}$$
$$= 2$$

$$\arg(\omega) = \pi - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$
$$= \pi - \frac{\pi}{6}$$
$$= \frac{5\pi}{6}$$



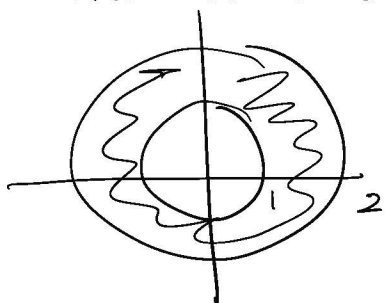
$\therefore \omega = 2 \operatorname{cis} \frac{5\pi}{6}$ 2 mark

(ii)

$$\omega^9 = \left(2 \operatorname{cis} \frac{5\pi}{6}\right)^9$$
$$= 2^9 \operatorname{cis} \left(\frac{5\pi}{6} \times 9\right)$$
$$= 512 \operatorname{cis} \left(\frac{15\pi}{2}\right)$$
$$= 512 \operatorname{cis} \left(\frac{3\pi}{2}\right)$$
$$= 512 \left[\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right]$$
$$= 512 [0 - i]$$
$$= -512i$$

1 mark

(b) Write the inequality that best describes the shaded area



$$1 < |z| \leq 2$$

1 mark

$$Q7(c) \quad z^3 + (1-5i)z^2 + (-5+2i)z + 15-3i = 0$$

$$i) \quad \bar{p} + \bar{q} + \bar{r} = \overline{p+q+r} \quad 1 \text{ mark}$$

$$= \overline{\left(-\frac{b}{a}\right)}$$

$$= \overline{-\left(\frac{1-5i}{1}\right)}$$

$$= \overline{-1+5i}$$

$$= -1-5i \quad 1 \text{ mark}$$

$$ii) \quad \bar{p} \times \bar{q} \times \bar{r} = \overline{p \times q \times r} \quad 1 \text{ mark}$$

$$= \overline{\left(-\frac{d}{a}\right)}$$

$$= \overline{-\left(\frac{15-3i}{1}\right)}$$

$$= \overline{-15+3i}$$

$$= -15-3i \quad 1 \text{ mark}$$

d) $z^6 = -1$

Find the first root

let a root be $z = r \cos \theta$

$\therefore z^6 = (r \cos \theta)^6 = -1$

$r^6 \cos 6\theta = -1$

$r^6 = 1$

$\cos 6\theta = \cos \pi$

$\therefore r = 1$

$6\theta = \pi$

$\theta = \frac{\pi}{6}$

$\therefore$ first root $z = \cos \frac{\pi}{6}$

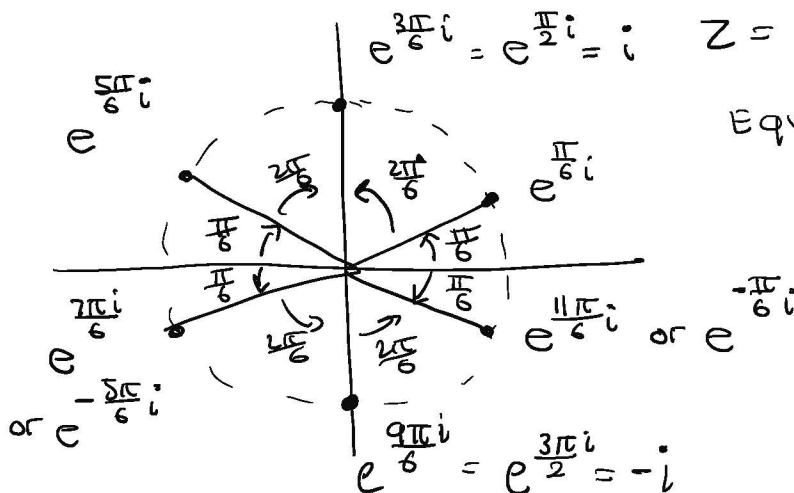
$e^{\frac{3\pi i}{6}} = e^{\frac{\pi i}{2}} = i$ $z = e^{\frac{\pi i}{6}}$ 1 mark.

Equal spacing $\frac{2\pi}{6} = \frac{\pi}{3}$

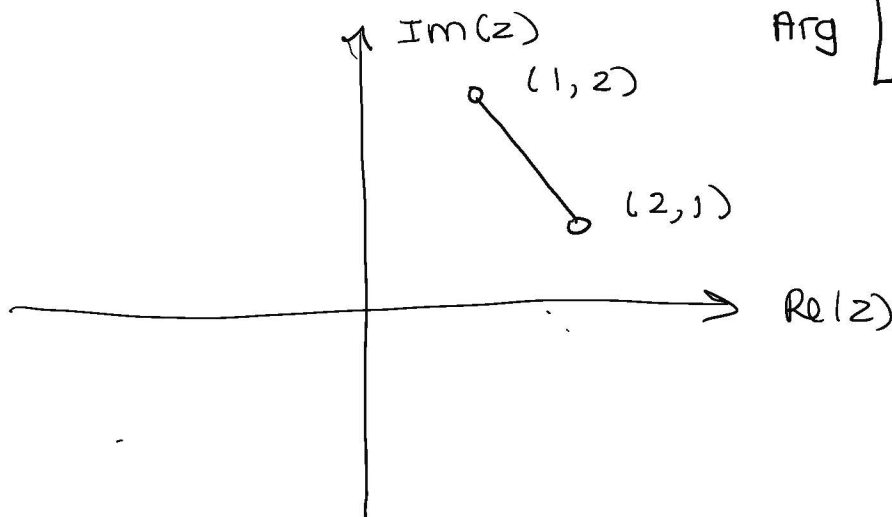
$\therefore$ the roots are

$e^{\frac{\pi i}{6}}, e^{\frac{2\pi i}{6}}, e^{\frac{3\pi i}{6}}, e^{\frac{4\pi i}{6}}, e^{\frac{5\pi i}{6}}, e^{\frac{6\pi i}{6}}$

1 mark for correct solution



e)



$\text{Arg} \left[\frac{z - (2+i)}{z - (1+2i)} \right] = \pi$

2 marks,

f) $\vec{OA} = r \cos \theta$ Rhombus so no scaling
 $\vec{OC} = \vec{OA} \times \cos\left(\frac{\pi}{6}\right) \times \cos\left(\frac{\pi}{6}\right)$ 1 mark.

$$= (1+4i) \times (\cos(\frac{\pi}{6} + \frac{\pi}{6}) + \sin(\frac{\pi}{6} + \frac{\pi}{6})i)$$

$$= (1+4i) \times (\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$$

$$= (1+4i) \times (\frac{1}{2} + \frac{\sqrt{3}}{2}i)$$

$$\vec{OC} = \frac{1}{2} + \frac{\sqrt{3}}{2}i + \frac{4}{2}i - \frac{4\sqrt{3}}{2}$$

ie $C = \frac{1}{2} - \frac{4\sqrt{3}}{2} + (\frac{4}{2} + \frac{\sqrt{3}}{2})i$

$\therefore \vec{OC} = \frac{1-4\sqrt{3}}{2} + (\frac{4+\sqrt{3}}{2})i$ 1 mark

$$\vec{OB} = \vec{OC} + \vec{OA}$$

$$= \frac{1-4\sqrt{3}}{2} + (\frac{4+\sqrt{3}}{2})i + 1+4i$$

$$\vec{OB} = \frac{1-4\sqrt{3}}{2} + (\frac{4+\sqrt{3}}{2})i + \frac{2+8i}{2}$$

ie $B = \frac{3-4\sqrt{3}}{2} + \frac{12+\sqrt{3}}{2}i$ 1 mark.

Q8

$$(a) \quad \text{Im} \left[\frac{1}{\bar{z} - i} \right] = 1$$

$$\text{let } z = x + iy$$

$$\bar{z} = x - iy$$

$$\begin{aligned} \frac{1}{\bar{z} - i} &= \frac{1}{x - iy - i} \\ &= \frac{1}{x - (y+1)i} \times \frac{x + (y+1)i}{x + (y+1)i} \\ &= \frac{x + (y+1)i}{x^2 + (y+1)^2} \end{aligned}$$

1 mark.

$$\rightarrow = \frac{x}{x^2 + (y+1)^2} + \frac{y+1}{x^2 + (y+1)^2} i$$

$$\text{Im} \left[\frac{1}{\bar{z} - i} \right] = \frac{y+1}{x^2 + (y+1)^2} = 1$$

$$y+1 = x^2 + y^2 + 2y + 1$$

$$0 = x^2 + y^2 + y$$

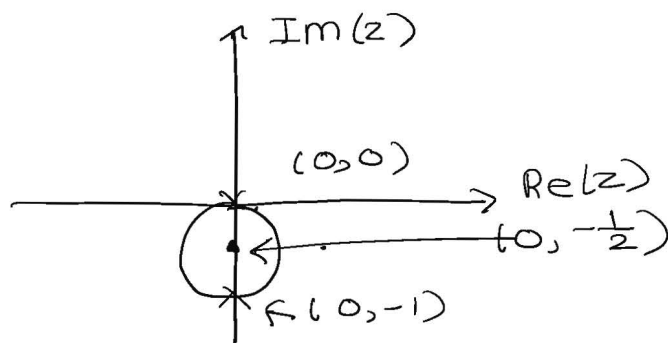
$$\left(\frac{1}{2}\right)^2 = x^2 + y^2 + y + \left(\frac{1}{2}\right)^2$$

$$\left(\frac{1}{2}\right)^2 = x + \left(y + \frac{1}{2}\right)^2$$

$$\frac{1}{4} = x + \left(y + \frac{1}{2}\right)^2$$

1 mark

$\therefore$ the locus is a circle at $(0, -\frac{1}{2})$ with a radius $\frac{1}{2}$.

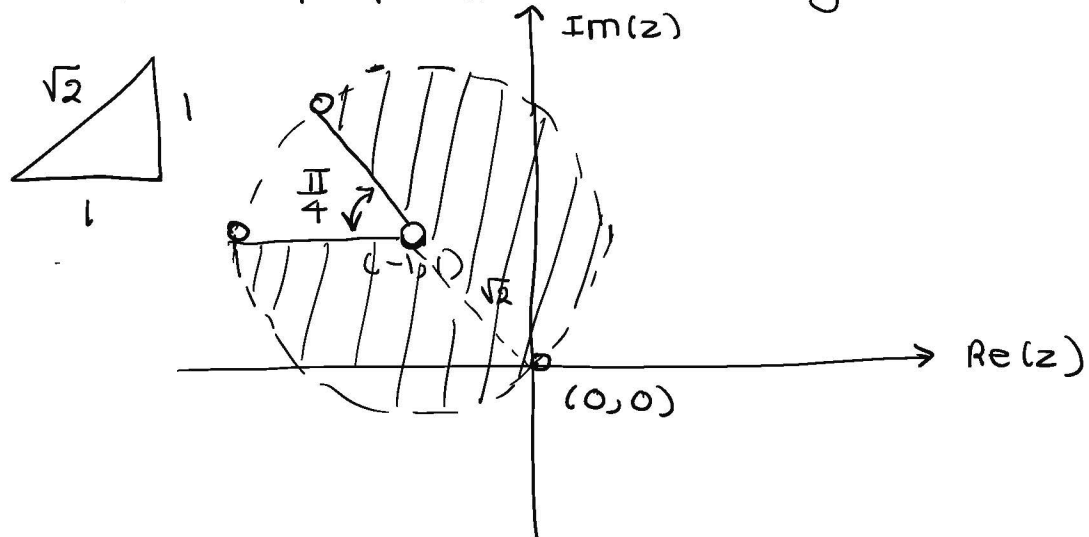


1 mark.

(b)

$$|z + 1 - i| < \sqrt{2} \quad \text{and} \quad -\pi \leq \arg(z + 1 - i) \leq \frac{3\pi}{4}$$

$$|z - (-1 + i)| < \sqrt{2} \quad \text{and} \quad -\pi \leq \arg(z - (-1 + i)) \leq \frac{3\pi}{4}$$



correct circle 1 mark.

correct angles 1 mark.

correct solution 1 mark.

c) $|z| < \frac{1}{3}$ show that $|(1+i)z^3 + iz| < \frac{4}{9}$

$$|(1+i)z^3 + iz| \leq |(1+i)z^3| + |iz| \quad \begin{array}{l} \text{Triangle} \\ \text{inequality} \\ (1 \text{ mark}) \end{array}$$

$$= |1+i||z^3| + |i||z|$$

$$= \sqrt{2}|z^3| + 1 \times |z|$$

$$< \sqrt{2} \left| \left(\frac{1}{3} \right) \right|^3 + 1 \times \left(\frac{1}{3} \right) \quad 1 \text{ mark}$$

$$= \frac{\sqrt{2}}{27} + \frac{1}{3}$$

$$= \frac{\sqrt{2}}{27} + \frac{9}{27}$$

$$< \frac{3+9}{27}$$

$$= \frac{12}{27}$$

$$= \frac{4}{9}$$

1 mark.

d) (i) $(\cos \theta + i \sin \theta)^5$

Binomial theorem to expand
 $(\cos \theta + i \sin \theta)^5 =$

$$\begin{aligned} & \binom{5}{0} \cos^5 \theta + \binom{5}{1} \cos^4 \theta \sin \theta i + \binom{5}{2} \cos^3 \theta \sin^2 \theta i^2 \\ & + \binom{5}{3} \cos^2 \theta \sin^3 \theta i^3 + \binom{5}{4} \cos \theta \sin^4 \theta i^4 \\ & + \binom{5}{5} \sin^5 \theta i^5 \end{aligned}$$

$$\begin{aligned} = & \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\ & - 10 \cos^2 \theta \sin^3 \theta i + 5 \cos \theta \sin^4 \theta \\ & + \sin^5 \theta i \end{aligned}$$

$$\begin{aligned} = & (\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta) \\ & + i(5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta) \end{aligned}$$

1 mark

(ii) Using de Moivre's theorem
 $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$

equating the imaginary parts

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

1 mark

(iii) equating the real parts

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta} \quad \text{1 mark}$$

$$= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

$$= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta} \div \frac{\cos^5 \theta}{\cos^5 \theta}$$

$$= \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

$$= \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{5 \tan^4 \theta - 10 \tan^2 \theta + 1} \quad 1 \text{ mark.}$$

□

(iv) $\tan 5\theta = 0$ when $\sin 5\theta = 0$

$$5\theta = k\pi \quad \text{where } k \in \mathbb{Z}$$

$$\theta = \frac{k\pi}{5}$$

$$\theta = 0, \pm \frac{\pi}{5}, \pm \frac{2\pi}{5}$$

Also $5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta = 0$

let $\tan \theta = x$

$$\therefore 5x - 10x^3 + x^5 = 0$$

$$x(x^4 - 10x^2 + 5) = 0$$

1 mark $\therefore x = 0 \quad x^4 - 10x^2 + 5 = 0$

$x = \tan \theta$ is the solution to $x = 0$

so the other solutions

$\tan(\pm \frac{\pi}{5}), \tan(\pm \frac{2\pi}{5})$ odd function

1 mark. $= \pm \tan \frac{\pi}{5}, \pm \tan(\frac{2\pi}{5})$ are also solutions